

KESTREL ROBOTICS · CAPABILITY STATEMENT

CORE COMPETENCIES

Autonomous ground robots for progress capture on active construction sites.

Visual-inertial SLAM that holds localization through airborne dust, changing site geometry and the absence of GPS. Automated comparison of as-built scans against the design BIM, with deviation reports generated overnight.

DIFFERENTIATORS

Our loop-closure approach tolerates a site whose geometry changes daily, which defeats conventional SLAM built on the assumption of a static map. Median drift under 4 centimetres over a 200 metre traverse on an active site.

PAST PERFORMANCE

Deployed on eleven commercial construction sites across three states. On a hospital expansion in Pittsburgh the system identified a 60 millimetre slab elevation deviation four weeks before it would have been found at inspection.

FACILITIES

Robotics laboratory and 6,000 square foot indoor test course in Pittsburgh, Pennsylvania, with a reconfigurable mock construction environment.